Exercise Sheet

# Lecture 5

## Eigenvalues and Eigenvectors

Throughout, the core uses a nonzero finite-dimensional $V$ over the field stated in the problem, $T \in \mathcal{L}(V)$ is an operator, and $E(\lambda, T) = \operatorname{null}(T - \lambda I)$ is its eigenspace. Here "spectrum" means the finite point spectrum; the general Hilbert-space spectrum is introduced in Lecture 14. Tags mark the flavour of each problem; a $\star$ marks a harder one.

### Core tutorial route

Work **5.1, 5.2, 5.7, 5.14, 5.18, 5.19, 5.20, and 5.21** in that order (about 80–100 minutes before discussion). This eight-problem route checks eigendata and characteristic invariants, real/complex transfer, an eigenspace proof, invariant decomposition and classification, reconstruction of the finite complex existence argument, and a constructive triangularizing flag.

**Additional practice:** 5.4–5.5, 5.9–5.13, 5.16–5.17, 5.22, 5.24. **Bridge:** 5.3, 5.6 (calculus and complex-field applications). **Extension:** 5.8, 5.15, 5.23 (harder counting/product proofs and generalized eigenvectors). Generalized-eigenvector work is not core progress.

### Finding eigenvalues and eigenvectors

**Exercise 5.1** (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 4 & -5 \\ 2 & -3 \end{pmatrix}$. Recover $\operatorname{tr} A$ and $\det A$ from the eigenvalues and check them directly.

Answer. Eigenvalues $2, -1$; $E(2, A) = \operatorname{span}\big((5, 2)\big)$, $E(-1, A) = \operatorname{span}\big((1, 1)\big)$; $\operatorname{tr} A = 1$, $\det A = -2$.

**Exercise 5.2** (computational) Compute the eigenvalues of the rotation matrix $\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}$ over $\mathbb{C}$. For which $\alpha$ are they real?

Answer. $\lambda = e^{\pm i\alpha}$; real iff $\sin\alpha = 0$, i.e. $\alpha \in \pi\mathbb{Z}$ (within $[0, 2\pi)$, just $\alpha = 0, \pi$).

**Exercise 5.3** (computational) Let $D \in \mathcal{L}(\mathcal{P}_2(\mathbb{R}))$ be differentiation. Find all eigenvalues and eigenvectors of $D$.

Answer. Only eigenvalue 0; $E(0, D) = \operatorname{span}(1)$ (geometric multiplicity 1, so $D$ is defective).

**Exercise 5.4** (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

Answer. Eigenvalues $3, -1$; $E(3, A) = \operatorname{span}\big((1, 1)\big)$, $E(-1, A) = \operatorname{span}\big((1, -1)\big)$.

**Exercise 5.5** (computational) Find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 0 & 1 \\ -6 & 5 \end{pmatrix}$.

Answer. Eigenvalues $2, 3$; $E(2, A) = \operatorname{span}\big((1, 2)\big)$, $E(3, A) = \operatorname{span}\big((1, 3)\big)$.

**Exercise 5.6** (computational) Working over $\mathbb{C}$, find the eigenvalues and a basis of each eigenspace for $A = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}$.

Answer. Eigenvalues $\pm i$; the eigenspaces are $E(i, A) = \operatorname{span}\big((1 + i, 1)\big)$ and $E(-i, A) = \operatorname{span}\big((1 - i, 1)\big)$.

**Exercise 5.7** (proof) Suppose $T$ is invertible and $\lambda$ is an eigenvalue of $T$. Prove that $\lambda \neq 0$ and that $1/\lambda$ is an eigenvalue of $T^{-1}$, with the same eigenvectors.

**Exercise 5.8** (⋆) **Extension.** Let $J$ be the $n \times n$ real matrix with every entry equal to 1, where $n \geq 2$. Find all eigenvalues of $J$, their algebraic and geometric multiplicities, and a basis of each eigenspace.

Hint. $J$ has rank 1; look at the all-ones vector and at the vectors whose entries sum to 0.

## Structure

**Exercise 5.9** (computational) Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Find the eigenvalues of $A^2 - A$.

Answer. 0 and 6.

**Exercise 5.10** (computational) Verify that $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is an involution and find $E(1, A)$ and $E(-1, A)$.

Answer. $A^2 = I$; $E(1, A) = \operatorname{span}\big((1, 1)\big)$, $E(-1, A) = \operatorname{span}\big((1, -1)\big)$.

**Exercise 5.11** (computational) The matrix $A = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$ is invertible. Find the eigenvalues of $A^{-1}$ together with a corresponding eigenvector for each.

Answer. Eigenvalues $\frac{1}{2}$ (eigenvector $(1, -1)$) and $\frac{1}{3}$ (eigenvector $(0, 1)$).

**Exercise 5.12** (proof) Suppose $\lambda$ is an eigenvalue of $T$ and $p \in \mathcal{P}(\mathbb{F})$. Prove that $p(\lambda)$ is an eigenvalue of $p(T)$, with the same eigenvector.

**Exercise 5.13** (proof) For finite-dimensional $V$, prove that $T$ is invertible if and only if 0 is not an eigenvalue of $T$.

**Exercise 5.14** (proof) Suppose $T^2 = I$ (an involution). Prove that every eigenvalue of $T$ is $\pm 1$, and that $V = E(1, T) \oplus E(-1, T)$.

Hint. For the eigenvalues, apply $T$ twice to an eigenvector. For the splitting, decompose $v$ into a part $T$ fixes and a part $T$ negates.

**Exercise 5.15** (⋆) **Extension.** Let $V$ be finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that $ST$ and $TS$ have the same eigenvalues.

Hint. If $STv = \lambda v$ with $\lambda \neq 0$, look at how $TS$ acts on $Tv$. Treat $\lambda = 0$ separately through invertibility.

**Exercise 5.16** (proof) Call $T$ *nilpotent* if $T^k = 0$ for some $k \geq 1$. Prove that the only eigenvalue of a nilpotent operator is 0.

## Upper-triangular form

**Exercise 5.17** (computational) Read off the eigenvalues of $A = \begin{pmatrix} 5 & 2 & 7 \\ 0 & 5 & -1 \\ 0 & 0 & 3 \end{pmatrix}$ and state its trace and determinant. Is $A$ invertible?

Answer. Eigenvalues 5, 5, 3; $\operatorname{tr} A = 13$, $\det A = 75$; invertible.

**Exercise 5.18** (proof) For $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$, classify all invariant subspaces among

$$U_1 = \operatorname{span}((1,0,0)), \qquad U_2 = \operatorname{span}((1,1,0)),$$
$$U_3 = \operatorname{span}((1,0,0),(0,1,0)), \quad U_4 = \operatorname{span}((0,0,1)).$$

Identify an invariant subspace that is not merely an eigenline.

Answer. $U_1, U_3, U_4$ are invariant; $U_2$ is not. The plane $U_3$ is invariant but is not an eigenline.

**Exercise 5.19** (proof) Let $T$ act on a nonzero complex vector space of dimension $n$. Reconstruct the finite eigenvalue-existence proof by explaining why $v, Tv, \ldots, T^n v$ is dependent, why the resulting annihilating polynomial has positive degree, where factorization over $\mathbb{C}$ is used, and why one linear factor must have nonzero kernel.

**Exercise 5.20** (computational) For $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ acting on $\mathbb{C}^2$, construct an ordered basis in which $A$ is upper triangular. Exhibit the corresponding invariant flag and verify the triangular matrix directly.

Answer. For example $v_1 = (1, i)$, $v_2 = (1, 0)$ gives $[A]_{(v_1,v_2)} = \left(\begin{smallmatrix} -i & -i \\ 0 & i \end{smallmatrix}\right)$; $0 \subset \operatorname{span}(v_1) \subset \mathbb{C}^2$ is invariant.

**Exercise 5.21** (proof) Suppose $T$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to a basis $v_1, \ldots, v_n$. Prove that $(T - \lambda_1 I) \cdots (T - \lambda_n I) = 0$.

**Exercise 5.22** ($\star$) Prove that if finite-dimensional $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then $T$ is diagonalizable (its eigenvectors form a basis).

Hint. Eigenvectors belonging to distinct eigenvalues are independent; count them against $\dim V$.

**Exercise 5.23** (computational) **Extension (generalized eigenvectors).** For $A = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$ acting on $\mathbb{F}^2$, find the eigenspace $E(3, A)$ and the generalized eigenspace $G(3, A)$, and exhibit a Jordan chain.

Answer. $E(3, A) = \operatorname{span}\big((1, 0)\big)$ (dim 1), $G(3, A) = \mathbb{F}^2$ (dim 2); Jordan chain $(0, 1) \to (1, 0) \to 0$.

**Exercise 5.24** (proof) Let $T \in \mathcal{L}(V)$ and $c \in \mathbb{F}$. Prove that $T$ and $T + cI$ have exactly the same eigenvectors, and that $\lambda$ is an eigenvalue of $T$ if and only if $\lambda + c$ is an eigenvalue of $T + cI$.